

SHEET METAL SHOP

PRACTICE-I

Time : 3 Hrs.

OBJECT:- Study of tools, operations and To make part 'A' funnel as per given drawing.

MATERIAL REQUIRED :- G.I. Sheet of 30 S.W.G. size 25cmx 25cm = 1 No.

TOOLS AND EQUIPMENTS USED:-

1) Measuring and Marking Tools:

- a) Engineering Scale
- b) Scriber
 - i) Straight Scriber
 - ii) Bent Scriber
- c) Divider
- d) Punch
 - i) Dot Punch
 - ii) Centre Punch
- e) Standard wire Gauge

2) Cutting Tools:

- a) Snips
 - i) Straight Snip
 - ii) Bent Snip

3) Supporting Tools:

- a) Bench Stakes
 - i) Bick Iron Stake
 - ii) Funnel Stake

4) Bending Tools:

- a) Hand Groover
- b) Straight Edge

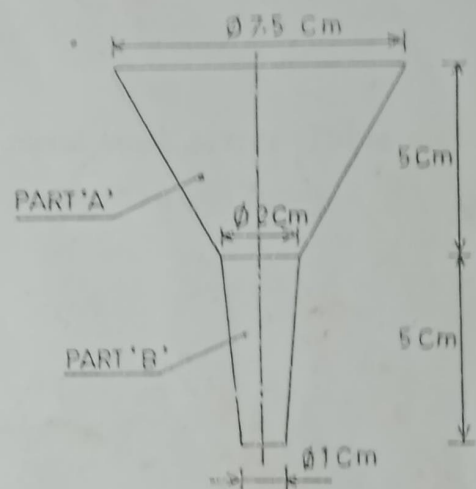
5) Striking Tools:

- a) Mallet
 - i) Round Head Mallet
 - ii) Rectangular Mallet

6) Soldering Tools:

- a) Soldering Iron

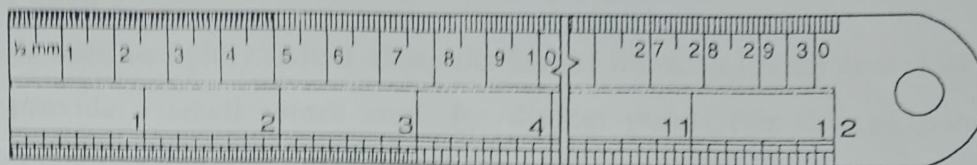
7) Smith's Furnace



DESCRIPTION OF TOOLS AND EQUIPMENTS

1. a) Engineering Scale :

It is a stiff, straight steel strip having all the faces machined true, on one of the flat faces graduation are marked in inches and centimeters. The least count is $1/64$ inch and 0.5mm

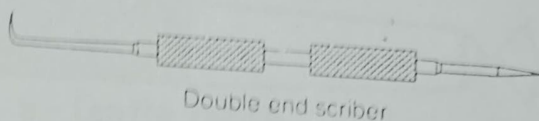


Engineering Scale

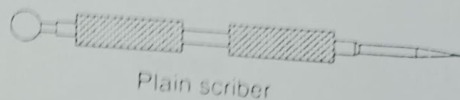
b) Scriber :

Scribers are used to scribe or mark lines on metal work pieces. These are made of High carbon steel. These are of two types as shown in figure.

- i) Straight Scriber
- ii) Bent scriber



Double end scriber

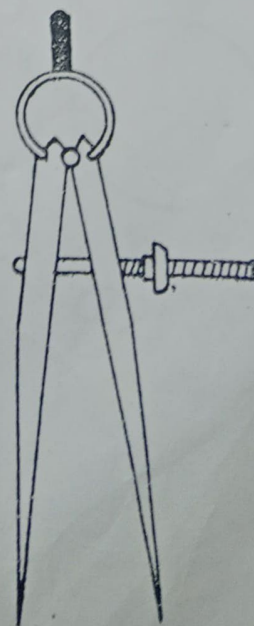


Plain scriber

Scriber

c) Divider :

The divider is used for drawing circles, and arcs on metal sheet. Two types are divider are available, in which spring divider is commonly used. Distance between legs of this divider are adjusted by turning the knurled nut.



d) Punch :

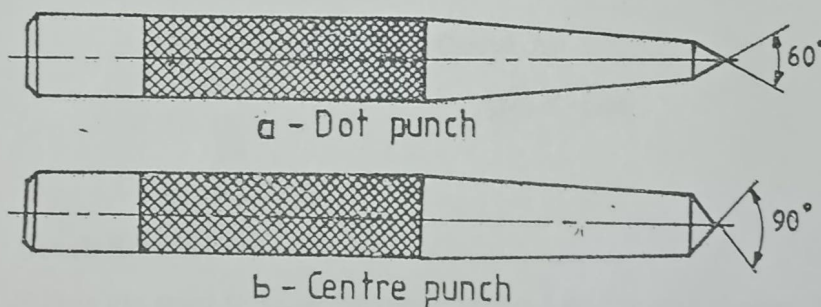
A punch is primary marking tool. It has either a serrated or knurled body to enable a firm grip or carries flat faces on its surface. One end of it is flat for hammering and the other end carries a point. It is used for marking round indentations on metal surface.

i) Dot Punch :

This is used to lightly indent along the layout lines, to locate centre of holes and provide a small centre mark for divider point. For this purpose, the punch is ground to a conical point having 60° included angle.

ii) Centre Punch :

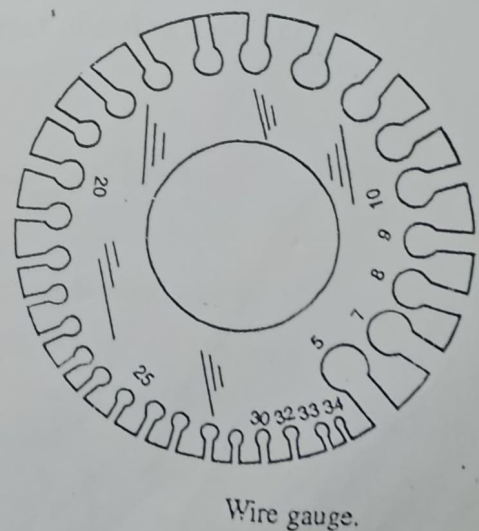
This is similar to dot punch, except that it is ground a conical point having 90° included angle. It is used to mark the location of holes to be drilled.



Punches

e) Standard Wire Gauge :

A standard wire gauge is a flat, circular, steel piece having slots all along its periphery, these slots have different standard size which are engraved near their bottom. The size of each slots represents the correct diameter of the wire of which its represents the gauge. The gauge number varies inversely as the size of wire. That is, the higher the gauge number the thinner the wire and vice versa. It is used to check the diameter of wire or thickness of sheet.



2. a) **Snips :**

Snips are used for cutting sheets to desired shapes and sizes.

- i) The straight snip are used for cutting along a straight line.
- ii) Bent snip are used for cutting the metal sheet along a curvature.



(a) Straight snip.



(b) Curved snip.

Types of snips.

3. a) **Bench Stakes :**

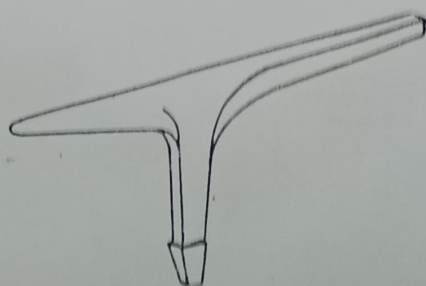
Stakes are used to form the metal sheets into various shapes. They actually work as a supporting tools as well as forming tools. They also help in bending operation. In operation they are fitted in the bench and then the job worked on them. These are available in different shapes and sizes to suit the requirement of the work. They are made of wrought Iron, Faced with steel. Different types of stakes used are:

i) **Bick Iron Stake :**

A bick iron stake is used in forming long tapered cylindrical items.

ii) **Funnel Stake :**

It has a tapered round working face and used for shaping conical work.



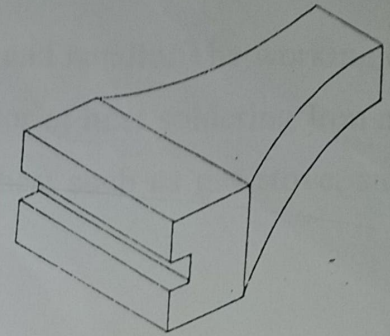
Bick iron



Funnel stake

4. a) Hand Groover :

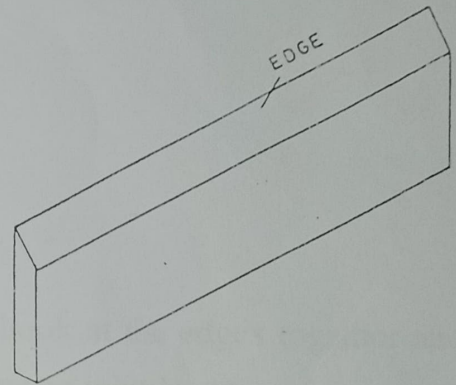
Hand Groover is used for locking the grooved seam. The outer ends of the groover should be slightly rounded to prevent them from marking the work. It is made in different sizes to suit the requirement.



HAND GROOVER

b) Straight Edge :

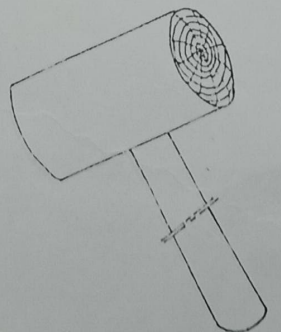
It has a long sharp straight edge bevelled along one side. It is used for making sharp bends, bending edges and forming pans, boxes etc. by hand.



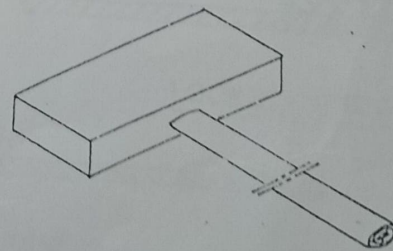
STRAIGHT EDGE

5. a) Mallet :

Mallets used in sheet metal work are made in various shapes and sizes to suit the different types of work. They are made to have round or square head to suit striking of the round surfaces and corner respectively. A wooden mallet is used in sheet metal work to avoid damage of sheets.



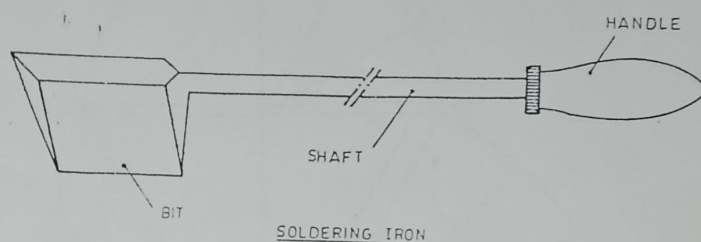
ROUND HEAD MALLET



RECTANGULAR MALLET

6. a) Soldering Iron :

It is used for soldering purpose. It consists of bit, shaft and handle. The working end of this tool is made of copper which is a good conductor of heat. Soldering iron may be heated electrically or by some external sources of heat such as gas stove, smith furnace etc.



TYPES OF JOINTS USED :-

a) Grooved joint:

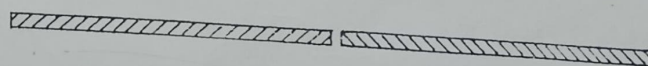
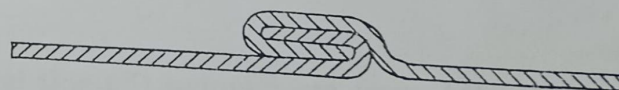
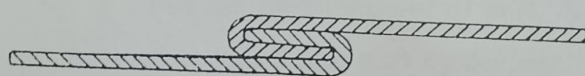
This joint is made by bending the sheet like a hook at the edges together and then setting the folding edges to make a joint. This type of joint is not very strong.

b) Locked grooved Joint:

It is a modified form of grooved joint. It is made by using a hand groover to lock the grooved joint so that they cannot be reopened.

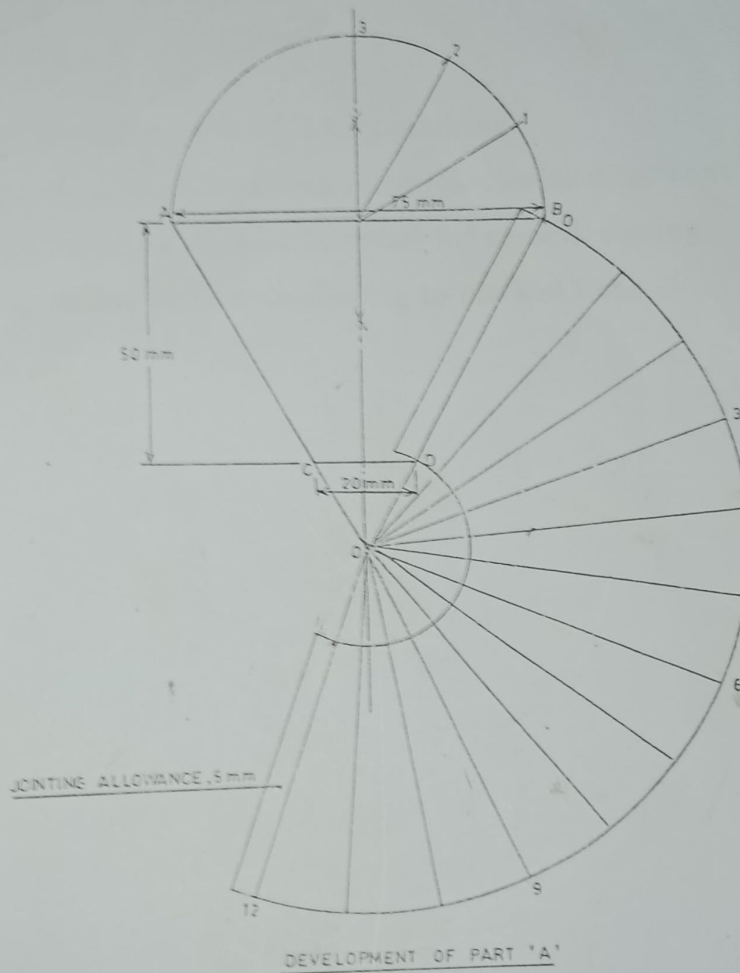
c) Butt Joint :

Both sheets are arranged edge to edge i.e. set parallel to each other and joined by soldering.



PROCEDURE :-

1. First divide the given figure of funnel into two parts i.e. part 'A' and part 'B'.
2. Then do the development of Part 'A' as shown in figure.



3. For the development of part 'A' draw a 7.5 cm horizontal line AB and divide it in two equal part by divider.
4. After dividation, draw a 2 cm horizontal line CD below 5 cm from AB and join point A to C and point B to D, which meet at point 'O's.
5. Next draw a semi circle on 7.5 cm line AB and divide it one Quadrant periphery into three equal parts.
6. Then take a diagonal distance OB & OD and Draw a arc by divider.
7. Take one equal part (1-2) and cut the 12 mark on the arc and join 0 to 12 no. and take 5 mm jointing allowance at both ends of sheet.

8. Thus the development is completed on the sheet as shown in figure. Now, cutting is done along the marking with snips and with the help of Funnel stakes. It is folded in the form of funnel by using grooved and locked grooved joint with hand groover.

PRECAUTIONS :-

1. Cutting should be smooth along marking.
2. Semi circle should be drawn in anticlockwise directions.
3. Concentration on work is essential for smooth cutting.
4. Cutting should be done according to marked lines.

PRACTICE-II

Time : 3 Hrs.

OBJECT:- To make part 'B' of funnel and complete the funnel with soldering.

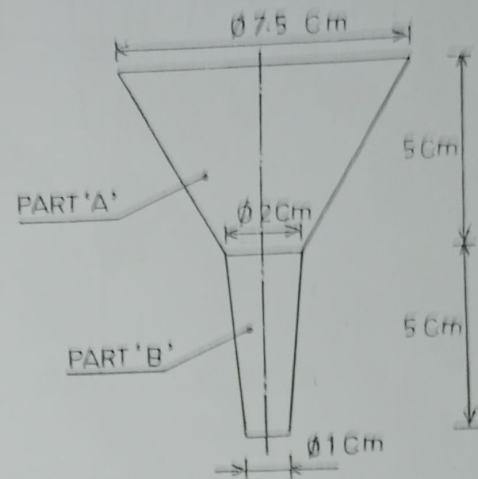
MATERIAL REQUIRED :- G.I. Sheet of 30 S.W.G. size 25cm x 25cm-01N6.

Solder - Sn 60%, Pb 40% Flux- ZnCl_2 , NH_4Cl

Steam coal

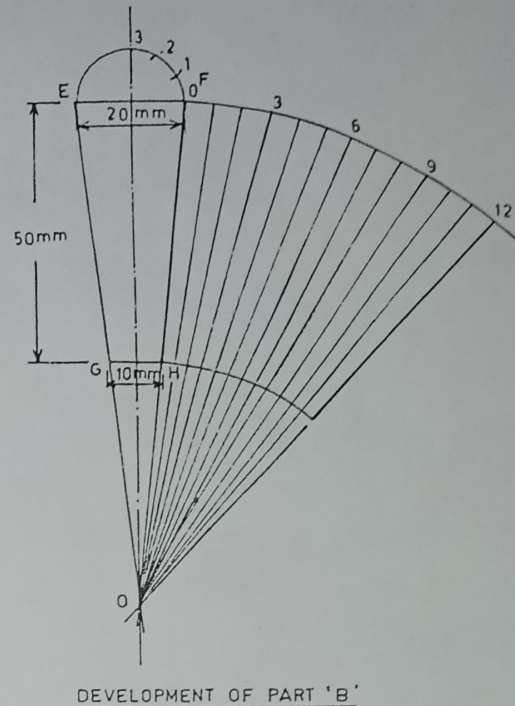
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- 4) Bending Tools:
 - a) Hand Groover
 - b) Straight Edge
- 5) Striking Tools:
 - a) Mallet
 - i) Round Head Mallet
 - ii) Rectangular Mallet
- 6) Soldering Tools:
 - a) Soldering Iron
- 7) Smith's Furnace:



PROCEDURE :-

1. For the development of part 'B', draw a 2 cm horizontal line EF and divide it in two equal parts by divider
2. After division, draw a 1 cm horizontal line GH below 5 cm from EF and join point G to E and point H to F, which meet at point 'O'
3. Next draw a semi circle on 2 cm line EF and divide its quadrant periphery into three equal parts.
4. Then take a diagonal distance OF and OH and draw an arc by divider.
5. Take one equal part (1-2) and cut the 12 mark on the arc and join O to 12 no.
6. Thus the development is completed on the sheet as shown in figure. Now cutting is done along the marking with snips and with the help of funnel stakes it is folded in the form of funnel by using butt joint.

**METHOD OF SOLDERING :-**

1. Clean the surfaces to be soldered.
2. After cleaning the parts, they are arranged in proper alignment and position.
3. Then the bit of soldering iron is heated, cleaned, dipped in flux and then rubbed on the solder to pick up a small quantity of solder with it and then deposited gently along the joint.

PRECAUTIONS :-

1. Cutting should be smooth along marking
2. Semi circle should be drawn in anticlockwise directions.
3. Concentration on work is essential for smooth cutting.
4. Cutting should be done according to marked lines.
5. Do not handle the soldered job, until the solder is partially cooled.

MASTER INDEX

(A+)

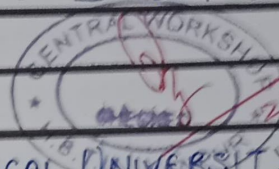
NAME: SHOBHIT DIXIT

CLASS: B-TECH. 1ST YEAR

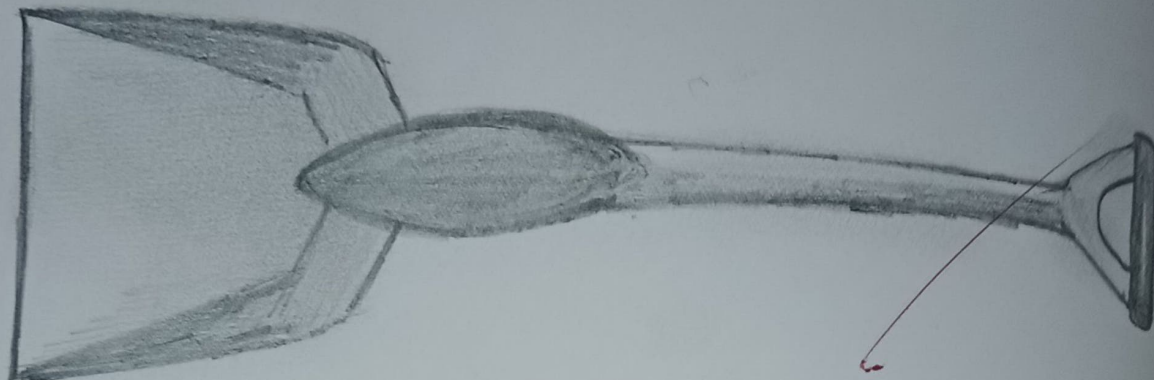
SR. NO: 19 PL 33

BRANCH: PLASTIC TECHNOLOGY

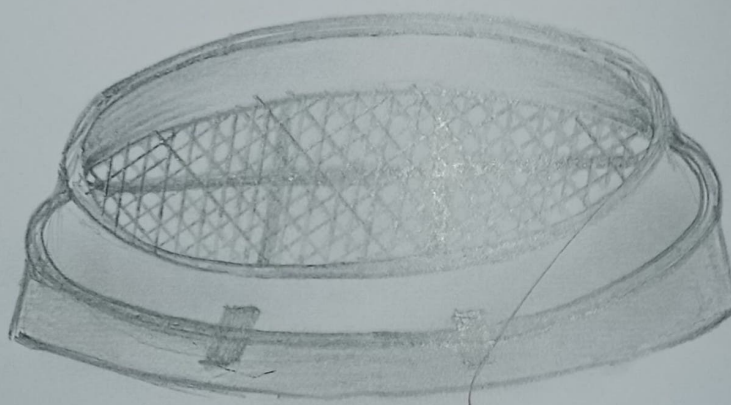
COLLEGE: HARCOURT BUTLER TECHNICAL UNIVERSITY



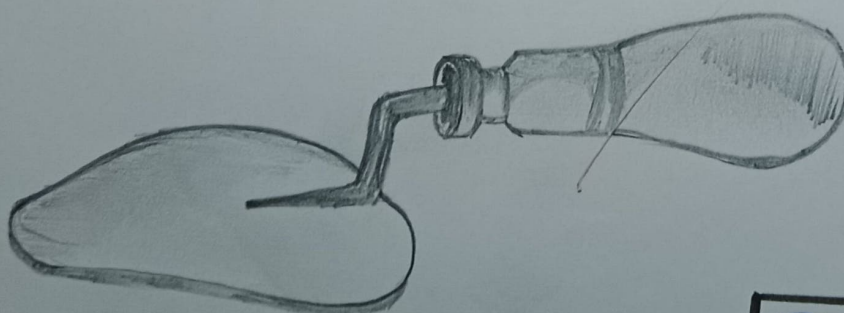
No.	Name of Shop	Practice No.	Date of Starting	Date of Finishing	Signature/Remarks
1.	FOUNDRY SHOP	1	26/08/19	26/08/19	Fair Verified 16/09/19
		2	02/09/19	02/09/19	
2.	WELDING SHOP	1	9/09/19	9/09/19	No 15/9/19
		2	16/09/19	16/09/19	
3.	MACHINE SHOP	1	30/09/19	30/09/19	?
		2	17/10/19	17/10/19	



SHOVEL



HAND-RIDDLE



TROWEL

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1st B.Tech
PLASTIC TECHNOLOGY

12/10/21